

OpenET Training: California Department of Water Resources

Location: CNRA new building 2-309, 715 P street, Sacramento, CA

Time: 8:30am - 4:30pm September 9-10, 2026

Instructors: Charles Morton (DRI); AJ Purdy (CSUMB); Alberto Guzman (CSUMB); Blake Minor (DRI); Forrest Melton (NASA ARC)

Get Set Up: Before the workshop, please install Python and the required dependencies by following the setup instructions in the [OpenET California DWR Technical Training GitHub repository](#). You will need Python installed, along with all packages listed in the repository's **requirements.txt** file.

Agenda

Day 1 — OpenET Fundamentals, Querying, Validation, and Applied Analysis

8:30–9:00 AM

Coffee, and Setup

9:00–9:20 AM

Welcome and Workshop Overview

Introductions, training goals, and an overview of OpenET.

9:20–10:15 AM

Python Basics for Working with OpenET Data Introduction to the Python tools and workflows used throughout the workshop.

10:15–10:30 AM

Break

10:30 AM–12:00 PM

API Query Basics: *Polygon Query Application*

Comparison with Ground-Based Measurements

Retrieve field-scale ET, create time series, match ground observations to measurement areas, and evaluate agreement between OpenET and field measurements.

12:00–1:00 PM

Lunch

1:00–2:15 PM

Harnessing the Field Summarized Data: *Using the OpenET Geodatabase endpoints to*

summarize ET by crop type: Compare ET within crop types across a region or among different crops within a region.

2:15–3:30 PM

Practical example: *Evaluating Conservation Program Impacts on Consumptive Use*

Explore treatment/control and before/after comparisons, baseline selection, attribution, and interpretation of program impacts.

3:30–3:35 PM

Break

3:35–4:20 PM

Review Day 1 workflows together

Hands-On Exercise: Apply method(s) from Day 1 to answer a question in groups.

One-on-one discussions

4:20–4:30 PM

Day 1 Wrap-Up

Questions, key takeaways, and preview of Day 2.

Day 2 — Scaling OpenET Analyses for Drought, Trends, and Consumptive Use

8:30–9:00 AM

Coffee and Optional Troubleshooting

Notebook setup, technical questions, and preparation for Day 2 exercises.

9:00–9:15 AM

Day 1 Recap and Day 2 Objectives

9:15–10:30 AM

Large Data Exports with Swagger / OpenET API

Explore API structure, authentication, batch requests, export strategies, and methods for handling larger areas and longer time periods. Includes raster composite exports and drought-impact comparisons. What to do? and What not to do?

10:30–10:45 AM

Break

10:45 AM–12:00 PM

OpenET Data in Google Earth Engine: Evaluating ET Trends Across California

Agricultural Lands & integrate OpenET with DWR land-use assets, create multi-year ET summaries, visualize trends, and compare regions.

12:00–1:00 PM

Lunch

1:00–2:00 PM

Consumptive Use with the Open Tool for Tracking Effective Precipitation. Combining ET and effective precipitation to evaluate consumptive use.

2:00–3:00 PM

Hands-on Exercise: Applying OpenET to a California Water-Management Question
Apply Day 2 method(s) from this workshop to answer a question.

3:00–3:05 PM

Break

3:05–4:00 PM

Group Discussion and Workshop Synthesis: Groups share analysis.

Review results, discuss approaches and interpretation, and identify opportunities for applying OpenET in DWR workflows.

4:00–4:15 PM

Closing Remarks and Training Evaluation